BACHELORTHESIS

In Biomedical Engineering

BOINSO – An open source solution for distributed ground and mission control communication for low earth orbit satellites

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Declaration

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Kurzfassung

In den letzten Jahren besteht die Möglichkeit für Forschungseinrichtungen und somit für die Lehre verwendete, kleine Satelliten zu bauen und in den Orbit zu schicken. Ermöglicht wurde dies durch die 1999 an der California Polytechnic State University (Cal Poly) entwickelten CubeSats.
Diese können für eine Vielzahl an Forschungsprojekten, jedoch auch für Telemedizinische Zwecke eingesetzt werden. So könnten beispielsweise Medizinische Daten wie zum Beispiel EKG-Signale mittels CubeSats von schwer erreichbaren Gegenden der Erde zu experten an weit entfernten Orten gesendet werden.
Da CubeSats sich im Low Earth Orbit (LEO) befinden, besteht teilweise nur ein 15 minütiges Zeitfenster in dem der Satellit das Mission Control Center (MCC) passiert, welches zu einer sehr geringen Datenausbeute führt [Preindl, 2010]. Um mehr Daten zu erhalten, wurde Global Educational Network for Satellite Operations (GENSO) entwickelt, ein weltweites Netzwerk aus Bodenstationen welche nun nichtmehr nur auf deren eigene CubeSats sondern, mit Berechtigung der MCCs, auch auf andere zugreifen können [Leveque et al., 2007].
Diese Arbeit beschäftigt sich mit der Entwicklung einer neuen Open Source Alternative, Berkley Open Infrastructure for Networking Satellite Operations (BOINSO), welche mithilfe von bereits vorhandenen Open Source Projekten und der Infrastruktur der Berkeley Open Infrastructure for Network Computing (BOINC) realisiert wird [Ries, 2012].
BOINSO wird unter der GNU General Public License (GNU GPL) vertrieben und steht unter https://github.com/Mehnen/BOINSO zum Download bereit.

Schlagworte: CubeSat, GENSO, BOINSO, Telemedicine

Abstract

During the last few years, there was a major increase in educational satellites launched to the orbit. The start of this was in 1999, when scientists at California Polytechnic State University (Cal Poly) developed the CubeSats.
The CubeSats can be used for a wide range of research projects, but also in the field of telemedicine.
Medical data like ECG could be sent from hard to reach places to experts located far away using the CubeSat.
Because of the CubeSats being in Low Earth Orbit (LEO), there is a very small timeframe for the satellites to send data to the MCC which leads to a very small data volume [Preindl, 2010].
To be able to get more data Global Educational Network for Satellite Operations (GENSO) was devised which connects a network of Ground Control Centers (GCCs) all over the world. Those GCCs can access the CubeSat of other teams and transmit the data of the passing satellite [Leveque et al., 2007].
This thesis focuses on the development of a new open source alternative, BOINSO, which is build using existing open source projects and the interface of BOINC [Ries, 2012].
BOINSO will be released under the GNU GPL and is available at https://github.com/Mehnen/BOINSO.

Keywords: CubeSat, GENSO, BOINSO, Telemedizin

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1 Introduction

Educational satellites like the CubeSats reside almost exclusively at the Low Earth Orbit (LEO), which ranges from 200 km to 2000 km above the earth surface, which has various reasons.
CubeSats are very small satellites and therefore the space for electronic devices and accumulators is very limited. Thus the short range between the device and GCCs is ideal as it reduces the power consumption to a minimum and is close enough to earth so that the CubeSat will not be affected by the radiation of the Van Allen radiation belts [Horne et al., 2005].
Moreover the deployment of a satellites in LEO significantly decreases the costs compared to satellites stationed in the Medium Earth Orbit (MEO) or Geostationary Orbit (GEO).
Furthermore, the closer the small satellites are to earth, the sooner they will burn to ashes caused by entering earth's atmosphere. This helps keeping the orbits clean, as space debris can cause collateral damage to future space missions [Akahoshi et al., 2008].
However, the small distance a satellite has in LEO leads to a severe problem to retrieve data, as data transmission is only possible when the CubeSat is visible to the GCC, leading to communication time-frames as short as 30 minutes. This fact, combined with a slow data rate makes a program like BOINSO indispensable.

2 Background

2.1 Global Educational Network for Satellite Operations (GENSO)

GENSO's aim is to provide a transmission network enabling a MCC to communicate with their respective educational satellites the whole day via GCCs located around the globe. This would prevent GCCs all over the world to reside unused for the major time of the day as normally every CubeSat sends to only one station.
Moreover, as the building of new educational satellites gets cheaper, more and more Universities will start their own "Space Program". These facilities will also be able to profit from the network as they don't neccessarily have to build a new GCC to gather the satellite's data.
To restrict the access to the satellites' data, each GCC has to register to an Authentication Server (AUS).

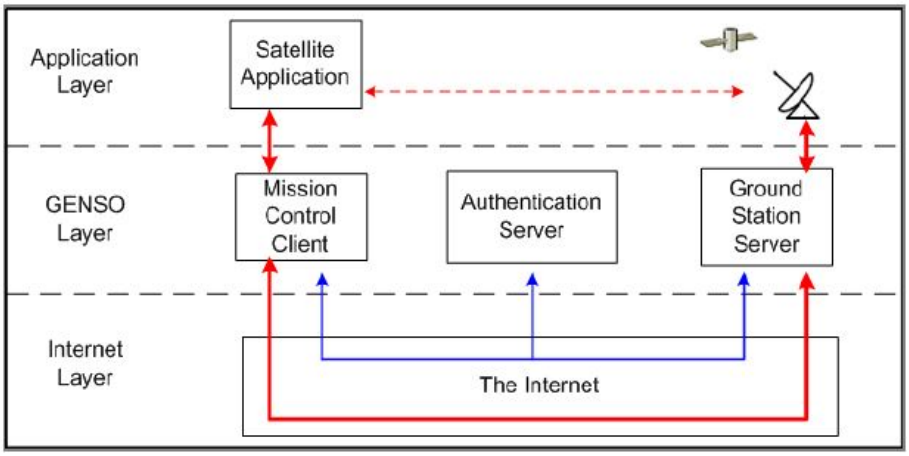

Figure 1: Layer Model of the GENSO System

2.2 CubeSats

The development of CubeSats started in 1999 with members of Cal Poly and Stanford University Space Systems Development Laboratory (SSDL). The nanosatellites' standard is $10 \times 10 \times 10$ cm with a maximum weight of 1.33 kg (1U CubeSat) [California Polytechnic State University, 2014].

Their main field of application are educational experiments conducted by Universities for example in -situ observations of lower thermospheric waves [Sanchez et al., 2007], but because to their being relatively low priced anyone who can raise enough money can build one. In 2012 there was even a successfully funded Kickstarter-project [ppl4world, 2012].
Another feasible field of use could be Telemedicine. CubeSats could be used to obtain and send telemetry data from hard to reach sources. Moreover, in countries with a low density of medical experts, the CubeSats could be used to send the data to experts located far away. There they could be analyzed and sent back.
In 2007 the *Universidad Distrital Francisco Jose de Caldas* proposed a plan to implement a telemedicine pilot project to transmitting and receiving ECG signals via CubeSats at *The 2007 CubeSat Developer's Workshop* [Pico and Vargas, 2007]. A project like that could lead to a much better healthcare for a lot of people in developing countries.
Several companies research the use of smartphone processors as the central avionics in in high altitude leading to the first successfull launch of a CubeSat stocked with the processor of the HTC Nexus One by Surrey Satellite Technology Ltd (SSTL) [Surrey Satellite Technology Ltd., 2011] and later that year to the PhoneSat project of the National Aeronautics and Space Administration (NASA) where they used the processors of the HTC Nexus One and Samsung Nexus S [Attai et al., 2014].

Figure 2: NASA PhoneSat - licensed under CC BY 2.0

3 Methods

To avoid further costs and proprietary software, there is the idea to build a satellite communication network consisting only of open source components.
The following pages show an assortment of open source projects that could be used to realized such an open source solution.

3.1 Berkeley Open Infrastructure for Network Computing (BOINC)

BOINC is a system to enable distributed computing and grid computing.
The organisation who wants other people to provide their computing power to them sets up a BOINC server and deploys a project on it. Now people who have the BOINC client installed from all around the world can register to the server and participate in projects they find interesting [Ries, 2012].
The basic idea behind this could also be used for a satellite network. Every MCC who wants to join the network, can set up a BOINC server containing the databases with the satellite informations. Now the GCCs with the BOINC client can connect to the server and poll for jobs. The MCC will then see which Ground Control Center sent the request as the client has to register first. Mission Control Center checks if the Ground Control Center meets all the requirements e.g. available bandwidth and resides at the right location for the satellite passing. Hereafter the MCC can decide whether to assign the GCC or not.
The Ground Control Center can then get the orbit data of the respective satellite from the NASA [**?**] and enter the received Two-line Element (TLE) in Gpredict (an introduction to Gpredict will be given in section 3.2) to adjust their antenna rotator.
If there is more than one CubeSat passing the GCC a scheduler has to be implemented to avoid conflicts and to choose the satellite according to the highest probability of high quality data [Preindl, 2010].
The received data will be stored at a local database and transmitted to the BOINC-server. After receiving the collected data the server can now crosscheck the results with information provided by other GCCs.

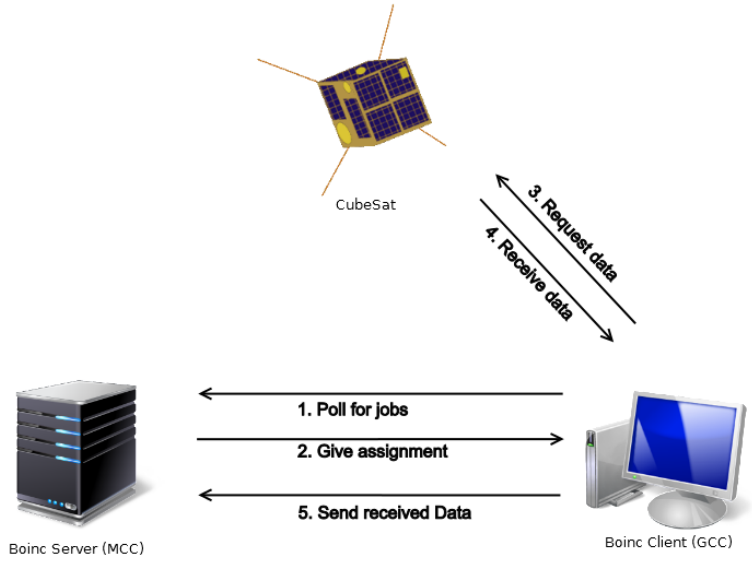

Figure 1: BOINC illustration

3.2 Gpredict

Gpredict is an open source program to track satellites and predict their respective orbits.
In general a satellite tracking program predicts the flight data of a satellite, like the position and velocity, by calculating values given by recieved Keplerian Elements, describing the orbit, using the *NORAD SGP4/SDP4* algorithm [Hoots and Roehrich, 1980]. After these values are retrieved other values like the bearing, distance and visibility can be calculated with the calculated orbit data and known location of the GCCs for a certain date and time. A general overview how a satellite tracking system works can be seen at 2. [Csete, 2011]

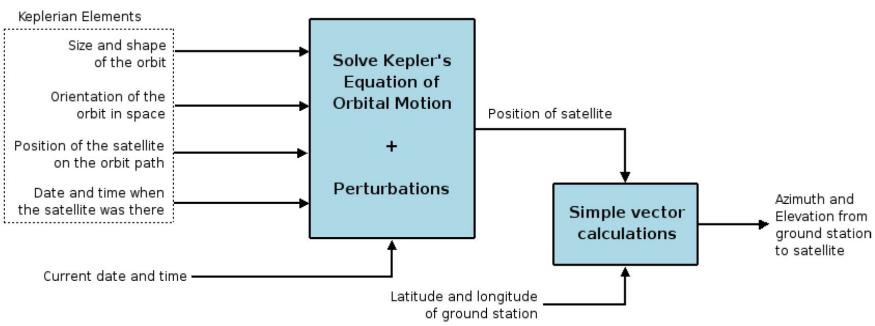

Figure 2: Functionality of a satellite tracking program - licensed under the GNU General Public License

3.3 GNU Radio

GNU Radio is a development toolkit to build your own software radio or signal processing application. The applications can be combined with external radio-frequency hardware or simulated. Due to the broad field of application GNU Radio is used by a lot of hobby radio operators, academic facilities and commercial environments [GNU Radio Website, b].
To build these applications, a graphical tool, GNU Radio Companion (GRC), is provided to draw a signal flowgraph.

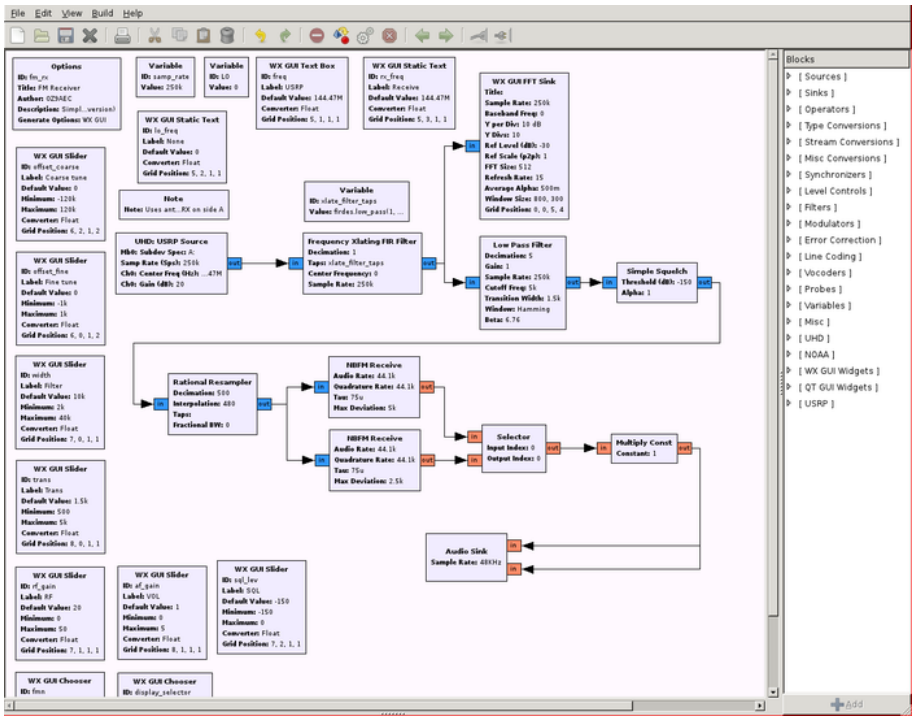

Figure 3: Editing a flowgraph in GNU Radio Companion - licensed under the GNU General Public License

One of the applications based on the GNU Radio is Fast Light Digital Modem Application (FLdigi). This or an adapted version like DI-FLdigi could be used to convert the data the GCCs receive from the MCC to the protocol the satellite uses.

4 Glossary

AUS Authentication Server

BOINC Berkeley Open Infrastructure for Network Computing

BOINSO Berkley Open Infrastructure for Networking Satellite Operations

Cal Poly California Polytechnic State University

cm Centimeter

FLdigi Fast Light Digital Modem Application

GCC Ground Control Center

GENSO Global Educational Network for Satellite Operations

GEO Geostationary Orbit

GNU GNU's Not Unix

GNU GPL GNU General Public License

GRC GNU Radio Companion

kg Kilogram

LEO Low Earth Orbit

MEO Medium Earth Orbit

MCC Mission Control Center

NORAD North American Aerospace Defense Command

NASA National Aeronautics and Space Administration

SSDL Stanford University Space Systems Development Laboratory

SSTL Surrey Satellite Technology Ltd

TLE Two-line Element

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